

# Emergence of Correlated Equilibrium Through Adaptive Learning: An Exposition of Foster and Vohra

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**Abstract**—Classical game theory traditionally studies equilibrium as the outcome of perfectly rational strategic reasoning. In many realistic environments, however, players repeatedly interact over time and adapt their behavior using only local information and past observations. This motivates the study of learning dynamics and the emergence of equilibrium behavior through repeated interaction.

The objective of this report is to examine the relationship between adaptive learning and correlated equilibrium by the means of the framework that was proposed by Foster and Vohra in 1997. We first begin by introducing the core intuition behind the correlated equilibrium and then we discuss how shared probabilistic signals enable richer forms of coordination in comparison to what is normally observed in the standard Nash equilibrium. We then discuss calibrated forecasting and repeated learning procedures in games, emphasizing how certain no-regret learning procedures can generate empirical play approaching correlated equilibrium in the long run.

To illustrate these ideas, we also study repeated adaptive play in the Shapley game, where persistent short-run oscillations coexist with increasingly stable long-run empirical distributions. While the players never settle on a single repeating action, the aggregate statistical behavior becomes considerably more regular over time. Finally, we discuss how the Foster–Vohra perspective extends naturally to decentralized multi-agent learning under uncertainty.

The goal of this report is to present the main ideas of the Foster–Vohra framework in an intuitive and accessible manner while emphasizing the connection between statistical learning and equilibrium emergence.

## I. INTRODUCTION

CLASSICAL game theory typically studies equilibrium as the outcome of perfectly rational strategic reasoning. In many practical settings, however, players repeatedly interact over time and gradually adapt their behavior using only local information and past observations. This motivates the study of learning dynamics in games and the emergence of equilibrium behavior through repeated interaction.

A natural question is whether equilibrium behavior can arise even when players never explicitly solve for an equilibrium. Classical forms of equilibrium such as Nash equilibrium make an assumption that players begin with mutually consistent expectations about each other’s behavior. While this assumption is mathematically convenient, it is less obvious from a practical perspective how such coordinated expectations would arise naturally in the case of repeated strategic interactions.

Foster and Vohra approach this problem using probabilistic forecasting and calibration. Instead of superficially assuming a

foresight of the equilibrium from the very beginning, players repeatedly create forecasts about the potential future actions of their opponents and then try to respond to those predicted forecasts adaptively. A key result of the paper is that when these forecasts are calibrated, the empirical distribution of play approaches correlated equilibrium.

This result is interesting both mathematically and conceptually because, It not only connects statistical learning with game theoretic equilibrium but also provides a dynamic interpretation of correlated equilibrium as something that can emerge through repeated decentralized interaction.

In this report, we will discuss the Foster–Vohra framework with significant emphasis on the underlying intuition and interpretation. We first introduce correlated equilibrium and calibrated forecasting through simple examples. We then proceed to examine how calibration eventually leads to correlated equilibrium in repeated games and also briefly discuss the broader relevance of these fundamental ideas in modern learning theory.

## II. PRELIMINARIES AND NOTATION

We begin by formalizing the game-theoretic setup. Brief interpretations are included where useful for intuition.

### A. Strategic Game Setup

We consider a repeated two-player game. For each player  $i \in \{1, 2\}$ , let  $S(i)$  denote a finite set of all possible pure strategies which is available to player  $i$ . Alternatively speaking,  $S(i)$  is the collection of all possible actions available to player  $i$  during the game.

The payoff obtained by a player  $i$  on playing the strategy profile

$$(s_1, s_2)$$

is denoted by:

$$u_i(s_1, s_2).$$

For the sake of simplicity, we are placing our primary focus on the two-player setting which was used in the Foster–Vohra exposition, although many of the underlying ideas do extend naturally to larger games.

### B. A Simple Coordination Example

The following example captures the basic mechanism. Assume that two players are to independently choose action  $A$  or action  $B$ . Both players will receive a payment of 1 if they choose the same action and a payment of 0 else.

|   | A      | B      |
|---|--------|--------|
| A | (1, 1) | (0, 0) |
| B | (0, 0) | (1, 1) |

In this game, successful coordination benefits both players. If both players somehow receive a common recommendation before choosing their actions, they may coordinate more effectively. This idea motivates the notion of a correlated strategy.

### C. Correlated Strategies

A correlated strategy is defined as a probability distribution that covers pairs of strategies:  $S(1) \times S(2)$ .

One may think of this as a random mechanism that recommends actions to players before the game is played.

An intuitive interpretation used in the paper is that of an “umpire” or mediator. First, a random signal is generated based on some probability distribution. Based on this signal, the umpire privately recommends an action to each player. Each player only observes their own recommendation.

For example, in the coordination game stated above, the umpire may decide to do a coin flip such that:

- if the outcome turns out to be Heads, both players are instructed to play  $A$ ,
- if the outcome turns out to be Tails, both players are instructed to play  $B$ .

If both players follow the recommendation, they always coordinate successfully and obtain payoff 1.

The key question is whether any player has an incentive to deviate after observing their recommendation. If in the end no player benefits from deviating, the correlated strategy forms a correlated equilibrium.

### D. Correlated Equilibrium

Formally, a correlated equilibrium is a correlated strategy under which no player can improve their expected payoff by deviating from the recommendation while assuming that the other players continue following theirs.

Conceptually, the idea is simple:

given the recommendation received, following the recommendation should be at least as good as deviating from it.

Unlike Nash equilibrium, correlated equilibrium permits players to choose actions based on a common random recommendation from a coordinator. This typically leads to a larger set of equilibria, many of which are easier to compute and analyze.

### E. Repeated Play and Forecasting

The main focus of Foster and Vohra is not merely the definition of correlated equilibrium, but rather how such equilibria may emerge dynamically through repeated play.

Suppose the game is played repeatedly over time. In every round  $t$ , the player 1 generates a prediction about the expected future behavior of the player 2. This forecasting scheme is denoted by

$f$ .

Thus the output of the forecasting rule at given time  $t$  could be written as

$$f(t) = (p_1(t), p_2(t), \dots, p_n(t)),$$

where

$$p_j(t)$$

represents the predicted probability that player 2 will choose strategy  $j$  at round  $t$ .

To track what actually happens during repeated play, define

$$X(j, t) = \begin{cases} 1, & \text{if player 2 plays strategy } j \text{ at time } t, \\ 0, & \text{else.} \end{cases}$$

Therefore,  $X(j, t)$  simply records whether strategy  $j$  was played at round  $t$ .

Next, let

$$N(p, t)$$

denote the number of rounds up to time  $t$  during which the forecasting rule produced the same forecast vector  $p$ . In other words,  $N(p, t)$  counts how often a particular forecast has appeared.

Finally, define

$$\rho(p, j, t) = \begin{cases} 0, & \text{if } N(p, t) = 0, \\ \frac{\sum_{s=1}^t \mathbf{1}_{f(s)=p} X(j, s)}{N(p, t)}, & \text{else.} \end{cases}$$

Although the notation appears technical at first, the interpretation is intuitive. The quantity  $\rho(p, j, t)$  measures the empirical frequency with which player 2 actually played strategy  $j$  among all rounds in which forecast  $p$  was issued.

This brings us to the idea of calibration. A forecasting rule is said to be calibrated if the predicted probabilities match the observed frequencies over a period of time. For example, if a forecasting scheme repeatedly predicts that a strategy will be played with probability (0.7), then approximately 70% of those rounds should indeed realize that strategy in the long run.

The connection established by Foster and Vohra is that calibrated forecasting procedures generate empirical play that approaches the set of correlated equilibria. Thus, equilibrium behavior does not emerge from explicit strategic computation, but instead by the means of repeated adaptive learning.

### III. WHY DOES CALIBRATION LEAD TO CORRELATED EQUILIBRIUM?

In the previous portion, we introduced notions of correlated equilibrium and calibrated forecasting. We now discuss the central contribution of Foster and Vohra: namely, that calibrated adaptive learning naturally generates correlated equilibrium in repeated games.

The main philosophical change in the paper is that equilibrium behavior is not taken for granted from the beginning. Instead, players interact over and over, make statistical predictions about how their opponents will act, and choose the best actions based on those predictions. Correlated equilibrium then develops as a limiting statistical feature of this ongoing interaction process.

#### A. Learning Through Forecasting

Suppose the game is played repeatedly over time. In each round  $t$  of the game, every player computes a probabilistic prediction about the future actions of the opponent. On the basis of this prediction, the player then chooses the best myopic (immediate) response, which is the action that will maximize the current expected payoff based on the predicted distribution.

Most importantly, players are not expected to have any amount of global equilibrium foresight. They do not calculate a Nash equilibrium directly, nor do they think recursively about the beliefs of other players. Instead, learning is local and adaptive:

- 1) Observe the previous play.
- 2) Make a forecast.
- 3) Choose the best response.
- 4) Update the forecast with new data.

A forecasting rule is said to be calibrated if the predicted probabilities match the observed frequencies over time. This means that if some forecasting method predicts that something is going to happen with a probability of  $p$ , then that thing should actually happen about  $p$  percent of the time in the long run.

For instance, if a player predicts that the opponent will choose strategy  $A$  with a probability of 0.7, then calibration means that strategy  $A$  should be used in about 70% of those rounds as the number of rounds increases.

Thus, calibration represents a weak but statistically meaningful form of learning consistency.

#### B. Empirical Distributions of Play

In the study of the asymptotic behaviour of the repeated game, Foster and Vohra consider the empirical distribution of play. Let,

$$D_t(x, y)$$

indicate the fraction of rounds up to time  $t$  where the player 1 chose a strategy  $x$  and the player 2 chose a strategy  $y$ .

Formally,

$$D_t(x, y) = \frac{\#\{s \leq t : (x_s, y_s) = (x, y)\}}{t}.$$

The quantity

$$D_t(x, y)$$

therefore describes the empirical frequency with which the outcome

$$(x, y)$$

has occurred during repeated play.

This distinction is important. Foster and Vohra do not study convergence of individual actions at each round. Rather, they study convergence of the long-run statistical behavior of the system. Even if players continue fluctuating between actions indefinitely, the empirical frequencies themselves may stabilize.

#### C. Intuition Behind the Main Result

The central theorem of the paper states that if players use calibrated forecasting rules and choose best responses to those forecasts, then every limit point of the empirical distribution of play is a correlated equilibrium.

The intuition behind this result is remarkably elegant.

Suppose that at some round a player chooses strategy  $x$ . Since the player is best responding to their forecast, the forecast distribution being used at that moment must be one under which  $x$  is optimal. Consequently, each observed action is chosen as a best response to the forecast available at that round.

Calibration now becomes the crucial bridge between forecasts and realized outcomes. Because the forecasting rule is calibrated, the empirical behavior of play becomes asymptotically consistent with the forecasts used to generate actions. In other words, the statistical behavior of the realized game becomes asymptotically consistent with the forecasts used to generate actions.

As a result, the empirical conditional distributions inherit the same best-response structure satisfied by the forecasts. In the limit, no player gains by systematically deviating from the observed statistical recommendation structure. This is precisely the defining property of correlated equilibrium.

Conceptually, the theorem demonstrates how global equilibrium structure can emerge from purely local adaptive behavior. Players do not explicitly compute a correlated equilibrium. Instead, equilibrium behavior appears organically through repeated forecasting and statistical learning.

#### D. Interpretation of the Result

One of the most important conceptual contributions of Foster and Vohra is the reinterpretation of equilibrium as an emergent phenomenon rather than a purely static solution concept.

Traditional Nash equilibrium analysis assumes that players somehow arrive at mutually consistent strategic expectations from the outset. In contrast, the Foster–Vohra framework replaces equilibrium foresight with adaptive statistical learning.

Under this perspective, correlated equilibrium arises naturally because calibrated players eventually develop statistically reliable beliefs regarding the behavior of others. Best responses to these calibrated forecasts then generate empirical

play that satisfies the consistency conditions of correlated equilibrium.

Thus, the paper provides a direct conceptual bridge between probabilistic learning and game theoretic equilibrium behavior.

#### IV. EXTENSIONS, LIMITATIONS, AND MODERN CONNECTIONS

The Foster–Vohra framework had a major impact on the study of learning in games. It moved the focus from static equilibrium computation to adaptive statistical behavior. In this section, we will discuss a few key ideas that resulted from the framework and how it connects to later developments in learning theory.

##### A. Correlated Equilibrium as a Learnable Object

One of the most important philosophical implications of the paper is that correlated equilibrium is not merely a static solution concept defined through fixed-point conditions. Instead, correlated equilibrium may emerge dynamically through repeated adaptive interaction.

The main theorem of Foster and Vohra shows that if players repeatedly form calibrated forecasts and choose myopic best responses to those forecasts, then the empirical distribution of play approaches the set of correlated equilibria. In this sense, equilibrium behavior is not imposed externally but instead appears as a statistical property of long-run interaction.

Moreover, the paper also suggests a deep relationship between calibrated forecasting and correlated equilibrium. Repeated calibrated learning procedures naturally generate empirical play approaching correlated equilibrium in the long run.

This viewpoint substantially changes the interpretation of equilibrium in game theory. Instead of viewing equilibrium as the outcome of perfect global reasoning, Foster and Vohra reinterpret it as an emergent statistical phenomenon which arises from decentralized adaptive learning.

##### B. Limitations of Fictitious Play

The importance of calibrated learning does become much clearer after comparison with earlier learning approaches like fictitious play.

In fictitious play, each player forms beliefs about the opponent’s future actions using empirical frequencies from previous rounds and then chooses a best response to those frequencies. Although the procedure appears intuitively reasonable, convergence is not guaranteed in general games.

A classical example of this limitation is the cyclic behavior observed by Lloyd Shapley. In certain repeated games, players continuously revise beliefs and best respond, yet the resulting play oscillates indefinitely instead of converging to a stable equilibrium.

Conceptually, the issue is that empirical frequencies alone need not produce statistically consistent forecasts. Players may continually react to outdated information, generating persistent cycles in behavior.

The Foster–Vohra framework addresses this issue by imposing calibration. Rather than just blindly tracking past

frequencies, calibrated forecasting enforces asymptotic consistency between predicted probabilities and realized empirical outcomes. This additional statistical consistency is precisely what allows correlated equilibrium behavior to emerge in the long run.

The distinction between the two approaches may be summarized informally as follows:

| <b>Fictitious Play</b>     | <b>Calibrated Learning</b>        |
|----------------------------|-----------------------------------|
| May cycle indefinitely     | Generates statistical consistency |
| Purely reactive updates    | Forecast-based adaptation         |
| Convergence not guaranteed | Approaches correlated equilibrium |

##### C. Connection to No-Regret Learning

The ideas introduced by Foster and Vohra later became closely connected to the theory of no-regret learning, which now forms a central component of modern algorithmic game theory and online learning.

Informally, a no-regret learning procedure ensures that over time, a player does not consistently perform worse than they would have if they had always played a fixed alternative strategy.

Certain no-regret learning procedures are closely connected to calibrated forecasting ideas and often generate empirical play approaching correlated equilibrium.

This relationship helps explain why correlated equilibrium plays such a natural role in repeated adaptive interaction. Even when players possess only local information and use simple decentralized update rules, long-run play may still organize itself into statistically stable equilibrium behavior.

Today, these ideas seem to appear in various different domains like online optimization, multi-agent learning, mechanism design, and algorithmic economics. The Foster–Vohra perspective can thus be otherwise thought of as an essential bridge linking classical equilibrium theory and modern learning based interpretations of strategic interaction.

#### V. ILLUSTRATIVE SIMULATION OF ADAPTIVE PLAY IN THE SHAPLEY GAME

To give more insight into the ideas from the previous sections, we conclude with a small simulation involving repeated adaptive interaction in the classical Shapley game. This experiment is not aimed at strictly following the exact forecasting method used by Foster and Vohra. Rather, it illustrates qualitatively similar statistical phenomena under adaptive repeated play. The simulation should therefore be interpreted as a conceptual illustration of adaptive statistical behavior rather than a direct numerical verification of the Foster–Vohra theorem. The goal is to showcase the differences between short-term fluctuations and long-term statistical trends. The Shapley game is interesting because it demonstrates that repeated best-response dynamics do not have to settle into a stable action profile. Instead, the game can continue cycling endlessly. However, as Foster and Vohra point out, equilibrium concepts based on observed distributions can still arise through adaptive learning.

### A. Simulation Setup

We consider a repeated game with two players having three possible actions each. The payoff matrices are essentially derived from the classic Shapley cyclic game as:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Player 1 receives a payoff of 1 when both players choose the same action, while player 2 on the other hand benefits only from cyclic mismatches:

$$1 \rightarrow 2, \quad 2 \rightarrow 3, \quad 3 \rightarrow 1.$$

In each round, every player is to make an empirical prediction about the future behavior of the opponent only by using the previously observed play. Based on this forecast, the player computes the expected payoffs and selects an approximate best response using a smooth probabilistic choice rule.

To avoid terminating in degenerate deterministic behavior, a small amount of random exploration is incorporated into the decision process by the means of stochastic action selection.

The interaction process mirrors the adaptive learning procedure discussed earlier in the paper:

- 1) observe previous play,
- 2) construct probabilistic forecasts,
- 3) choose a best response,
- 4) update empirical beliefs using observed outcomes.

We run the simulation for several thousand rounds to try and observe how the empirical behavior evolves over time.

### B. Rolling Action Frequencies

Figure VI displays rolling empirical frequencies of the three actions during repeated play.

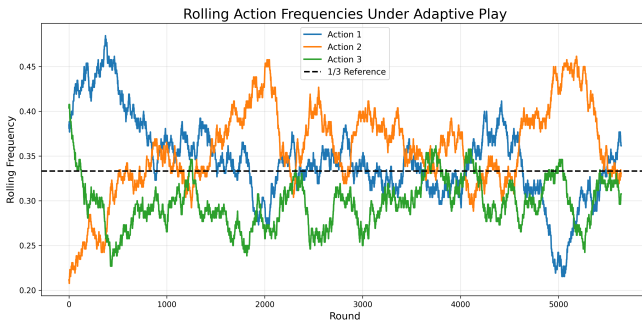


Fig. 1. Rolling action frequencies under adaptive play in the Shapley game. The dashed line indicates the symmetric benchmark frequency of  $1/3$ .

The figure illustrates the persistent oscillatory nature of adaptive play in the Shapley game. No single action stabilizes permanently, and the players continue fluctuating between competing responses throughout the repeated interaction process.

At the same time, the rolling frequencies remain centered around the symmetric benchmark value of

$$\frac{1}{3}.$$

Thus, even though local play still remains dynamically unstable, the system begins to exhibit some form of long-term statistical balance. This distinction between short-run oscillation and long-run regularity is one of the central conceptual themes underlying the Foster–Vohra framework.

### C. Empirical Joint Distributions

To understand in depth the long-term statistical structure of play, we also examine the empirical frequencies of several joint action profiles.

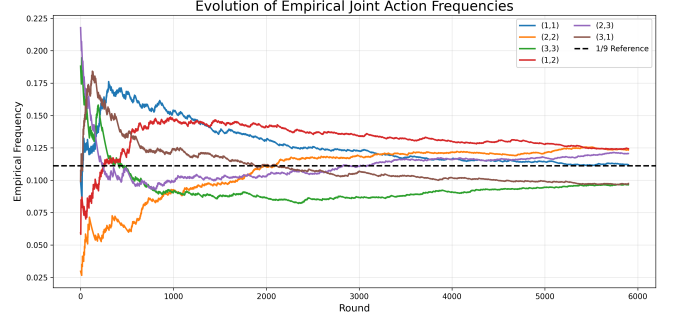


Fig. 2. Evolution of empirical joint action frequencies during repeated adaptive interaction. The dashed line indicates the uniform benchmark frequency  $1/9$ .

In contrast to the rolling action frequencies, the empirical joint distributions exhibit much more long-run stability. Empirical frequencies tend to stabilize over time, although there are transient fluctuations in the early stages of the simulation.

This behavior reflects the key philosophical insight of Foster and Vohra. Individual actions need not converge pointwise to a fixed equilibrium strategy. Instead, equilibrium behavior may emerge statistically through the empirical distribution generated by repeated adaptive interaction.

In particular, the simulation depicts how persistent local oscillations may still coexist with increasingly regular long-run empirical structure. This phenomenon motivates the interpretation of correlated equilibrium as a statistical property which emerges from adaptive learning rather than explicit equilibrium computation.

### D. Interpretation

The simulation highlights an important conceptual distinction between action-level convergence and distributional convergence.

The Shapley game demonstrates that repeated best-response behavior may continue cycling indefinitely. Regardless, the empirical distribution of play becomes more and more structured as time progresses. This aligns closely with the Foster–Vohra perspective that equilibrium should be interpreted statistically through long-run empirical behavior rather than through instantaneous stabilization of actions.

It is important to emphasize that the experiment is purely pedagogical and does not constitute a rigorous implementation of calibrated forecasting algorithms. Rather, the simulation is intended to provide intuition for the broader philosophical message of the paper:

equilibrium-like statistical regularity may emerge organically through decentralized adaptive learning even when individual actions continue fluctuating indefinitely.

## VI. A MULTI-AGENT INTERPRETATION THROUGH SOCIAL DEDUCTION GAMES

Although Foster and Vohra present their exposition primarily in the two-player setting, the conceptual framework underlying calibrated learning extends naturally to more complex multi-agent environments. In this section, we discuss how the philosophy of the paper may be interpreted through social deduction games such as *Mafia*. The purpose of this discussion is not to rigorously analyze Mafia from a formal game theoretic lens, but rather to use it as an intuitive case study showcasing repeated probabilistic forecasting, adaptive belief formation, and decentralized strategic coordination.



### A. Repeated Strategic Interaction Under Uncertainty

Social deduction games create a unique setting for studying adaptive learning because players work with incomplete information. They constantly update their beliefs based on what they observe.

Imagine a game with  $n$  players. One or more of them secretly belong to a mafia group, while the rest are innocent. In each round, players try to talk openly, share their suspicions with others, and ultimately end up in a vote to eliminate one player from the game at the end decided by majority opinion. The identities of players are only partly visible, and success relies heavily on reasoning under uncertainty.

Unlike traditional static games, players in Mafia do not calculate equilibrium strategies. Instead, they continuously adjust their beliefs based on the evidence they gather such as:

- 1) previous voting patterns,
- 2) public accusations,
- 3) revealed eliminations,
- 4) consistency of player behavior,
- 5) historical outcomes from prior games.

This creates a repeated forecasting environment remarkably similar in spirit to the Foster–Vohra framework.

### B. Forecasting and Strategic Updates

Suppose that player  $i$  assigns a probability

$$p_i^{(t)}(j)$$

to the event that player  $j$  is secretly mafia at round  $t$ .

These probabilities evolve dynamically with repeated interaction. For example, if player  $j$  repeatedly shows up in suspicious voting coalitions or acts in a way that contradicts what was said before, player  $i$  might increase the associated forecast probability.

At every round, players then choose actions based on these forecasts. A player may vote against the individual assigned the highest probability of being mafia, or strategically coordinate with other players whose forecasts appear reliable.

Thus, the interaction process follows the same broad adaptive structure emphasized throughout Foster and Vohra:

- 1) observe historical play,
- 2) form probabilistic forecasts,
- 3) choose a myopic strategic response,
- 4) update beliefs after observing outcomes.

Most importantly, no player is expected to have complete global information about the game's strategy at any time. Coordination is instead expected to develop gradually through decentralized statistical learning.

### C. Calibration Interpretation

The notion of calibration appears naturally in this setting.

Suppose that over many repeated games, player  $i$  repeatedly assigns a probability of approximately 0.8 to the claim that player  $j$  is mafia. If player  $j$  is mafia in roughly 80% of the situations in which player  $i$  assigns probability 0.8, then the forecasting rule used by player  $i$  may be interpreted as approximately calibrated.

In this sense, calibration measures the long-run statistical consistency between predicted strategic beliefs and realized outcomes.

This is important because social deduction games involve uncertainty, hidden information and probabilistic reasoning. Hidden roles are not directly visible to players. They have to make do with gathering evidence over the long term and statistical reasoning.

Consequently, the Foster–Vohra viewpoint suggests a broader interpretation of equilibrium behavior:

coordinated strategic behavior may emerge not through explicit equilibrium computation, but through repeated probabilistic forecasting and adaptive learning.

### D. Extension Beyond the Two-Player Setting

The Mafia interpretation also helps illustrate why the ideas of Foster and Vohra naturally extend beyond the two-player framework used in the original exposition.

In the case of two players, the forecasts are designed to predict the future actions of one of the opponents. However, in a larger multi-agent environment, players need to consider the behavior and motivations of multiple interacting agents at the same time.

But the central insight identified by Foster and Vohra remains conceptually unchanged. Players make probabilistic forecasts, update their beliefs based on observed outcomes,

and adjust as necessary. This decentralized adaptive process leads to increasingly structured statistical behavior over time.

Thus, although the formal exposition in the paper focuses on two-player repeated games for analytical simplicity, the broader philosophy of calibrated adaptive learning naturally extends to decentralized multi-agent systems involving uncertainty and incomplete information.

#### E. Connections to Learning in Multi-Agent Systems

The Mafia interpretation also highlights why the Foster–Vohra framework remains conceptually relevant in modern multi-agent environments.

In strategic games that are repeated, the players often do not know completely the intentions or internal states of the other players. Instead, they learn from the interaction data they have seen so far, and try to adapt their behavior accordingly. This is the decentralized statistical learning procedure considered in Foster and Vohra .

Similar ideas are now seen in several new modern settings like:

- online learning systems,
- decentralized markets,
- multi-agent reinforcement learning.

In all of these environments, agents continually update beliefs using empirical observations and choose strategies based on those evolving forecasts.

The central insight of Foster and Vohra is therefore broader than the specific calibrated forecasting procedure studied in the paper. The framework demonstrates that equilibrium-like statistical regularity may emerge even when agents possess only local information and repeatedly adapt using observed outcomes.

This view treats correlated equilibrium not as a static mathematical object, but as a natural statistical structure that emerges from decentralized adaptive interaction.

## VII. CONCLUSION

The work of Foster and Vohra provides a new and unique perspective on player behavior in repeated games. Instead of simply just assuming that players have full strategic planning to back up their moves or assuming that players calculate their strategies perfectly(or ideally), the framework demonstrates how equilibrium-consistent statistical behavior can arise through repeated forecasting, observation, and learning.

Our main contribution is to connect calibration to correlated equilibrium. A calibrated forecast is one where the predicted probability matches the observed outcome. These forecasts are transformed into strategic decisions through myopic best responses. These processes combine to produce empirical play that approaches the set of correlated equilibria.

More generally, the view of Foster and Vohra led to a learning approach to game theory. This approach quickly became a key ingredient in several fields such as no-regret learning, online optimization, and multi-agent systems. The paper is important for the development of current algorithmic and learning approaches to strategic interaction.

As a summary, the Foster and Vohra framework demonstrates that equilibrium need not be viewed solely as a product of perfect player-level rationality. Instead, we can think of it as something that can naturally arise as a statistical feature of continuous interactions and learning.

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